

NEWARK INTL, NJ, USA

WMO: 725020

Lat: 40.683N

Lon: 74.169W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14734

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.8	-21.2	0.6	-9.0	-18.9	0.7	-6.5	12.7	-1.4	11.8	-1.1	6.2	280	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	34.5	23.5	32.8	22.7	31.3	22.2	25.3	31.2	24.5	29.7	23.8	28.5	5.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.5	27.7	22.9	17.7	26.9	22.3	17.0	26.5	77.0	31.2	73.7	29.9	71.0	28.3	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.9	8.7	-13.9	37.3	2.9	1.8	-15.9	38.6	-17.6	39.7	-19.2	40.7	-21.3	42.1		
(5)			-15.1	26.5	2.7	1.1	-17.0	27.4	-18.5	28.0	-20.0	28.7	-21.9	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	0.4	1.8	6.0	12.1	17.4	22.6	25.6	24.7	20.9	14.4	8.4	3.5		
	DBStd	9.76	5.59	5.14	4.99	4.58	4.17	3.67	2.95	2.78	3.51	4.22	4.54	4.85		
	HDD10.0	1006	302	233	144	27	1	0	0	0	0	9	82	208		
	HDD18.3	2581	557	463	384	196	69	7	0	1	15	134	297	459		
	CDD10.0	2181	3	3	19	89	231	379	485	456	326	147	36	8		
	CDD18.3	714	0	0	1	8	40	137	227	198	91	13	1	0		
	CDH23.3	6276	0	1	9	88	375	1200	2237	1703	588	73	2	0		
	CDH26.7	2307	0	0	1	25	128	446	934	602	157	13	0	0		
(14) Wind	WSAvg	4.3	4.8	4.8	5.0	4.7	4.1	4.0	3.9	3.7	3.9	4.2	4.4	4.6		
(15) Precipitation	PrecAvg	1110	85	74	102	94	104	86	111	102	99	73	92	88		
	PrecMax	1664	256	126	283	283	260	267	254	270	238	172	293	240		
	PrecMin	663	11	13	28	23	13	10	23	13	26	5	13	16		
	PrecStd	213	51	29	50	50	54	49	63	55	55	39	59	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	18.5	24.2	30.1	33.0	35.5	37.1	35.7	33.1	28.5	22.2	19.3		
		MCWB	14.0	12.7	16.0	18.3	21.1	23.8	24.1	24.3	22.0	19.7	16.6	15.6		
	2%	DB	13.8	14.5	19.4	25.9	30.2	33.0	34.9	33.4	30.3	25.3	19.6	15.6		
		MCWB	11.6	10.1	12.4	15.9	20.2	22.9	23.6	23.3	21.6	18.8	15.2	12.7		
	5%	DB	10.4	11.5	16.0	22.5	27.6	31.1	33.1	31.6	28.3	23.1	17.5	12.9		
		MCWB	7.8	7.3	10.2	14.3	18.7	21.9	22.9	22.6	20.8	17.8	13.6	10.4		
	10%	DB	8.0	8.9	13.4	19.7	24.9	29.2	31.4	29.9	26.6	21.2	15.5	10.6		
		MCWB	5.5	5.4	8.7	12.4	17.4	20.6	22.5	21.7	20.1	16.6	12.2	7.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	15.0	16.8	19.7	22.9	25.4	26.5	26.1	24.7	22.2	18.6	16.2		
		MCDB	16.6	17.6	22.3	26.8	29.9	32.5	33.9	32.7	29.3	24.9	20.6	18.3		
	2%	WB	12.0	10.9	14.0	17.6	21.5	24.1	25.5	25.1	23.8	20.7	16.5	13.3		
		MCDB	13.2	13.8	17.5	23.7	28.0	30.2	31.7	30.5	27.2	23.8	18.7	15.2		
	5%	WB	8.1	7.9	11.3	15.7	20.3	23.1	24.7	24.3	22.8	19.1	14.5	11.0		
		MCDB	9.9	10.5	14.9	20.9	25.7	28.8	30.1	28.7	26.1	22.1	16.5	12.7		
	10%	WB	5.7	5.8	9.1	13.7	18.7	22.2	23.9	23.5	21.7	17.5	12.7	8.1		
		MCDB	7.7	8.5	12.7	18.0	23.5	27.4	28.9	27.7	25.1	20.4	14.9	9.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.0	8.6	9.5	9.5	9.1	8.9	8.5	8.6	8.6	8.0	6.9		
		MCDBR	10.4	11.5	12.4	13.7	13.2	11.6	11.1	10.5	10.6	10.8	10.3	9.8		
	5% WB	MCWBR	9.0	8.6	8.0	7.0	6.1	4.8	3.9	4.1	4.7	5.9	7.4	8.6		
		MCWBR	9.9	10.6	11.5	12.2	11.8	10.2	9.4	8.7	8.1	8.9	8.8	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.328	0.362	0.385	0.432	0.493	0.494	0.481	0.403	0.364	0.332	0.314		
	taud		2.459	2.422	2.389	2.357	2.256	2.124	2.135	2.163	2.352	2.473	2.518	2.521		
	Ebn at Noon		844	885	888	887	849	795	790	788	833	834	819	818		
	Edh at Noon		78	93	108	119	135	155	151	143	111	89	74	69		
(44) All-Sky Solar Radiation	RadAvg		1.87	2.77	3.72	4.79	5.31	5.83	5.95	5.16	4.30	3.00	2.11	1.59		
	RadStd		0.16	0.28	0.31	0.47	0.56	0.53	0.29	0.34	0.43	0.30	0.18	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	+68

Nomenclature: See separate page